

Lab 8

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Deadline: 2022-11-28 22:30:00 Beijing Time

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Exercise09_01.java

(The Rectangle class) Design a class named Rectangle to represent a rectangle. The class contains:

- Two double data fields named width and height that specify the width and height of the rectangle. The default values are 1 for both width and height.
- A no-arg constructor that creates a default rectangle.
- A constructor that creates a rectangle with the specified width and height.
- A method named getArea() that returns the area of this rectangle.
- A method named getPerimeter() that returns the perimeter.

Exercise09_01.java

Write a test program that creates two Rectangle objects—one with width 4 and height 40, and the other with width 3.5 and height 35.9. Display the width, height, area, and perimeter of each rectangle in this order:

```
The area of a rectangle with width 4.0 and height 40.0 is 160.0
The perimeter of a rectangle is 88.0
The area of a rectangle with width 3.5 and height 35.9 is 125.64999999999999
The perimeter of a rectangle is 78.8
```

Exercise09_04.java

(Use the Random class) Write a program that creates a Random object with seed 1000 and displays the first 50 random integers between 0 and 100 using the `nextInt(100)` method.

Here is a sample run of the program:

```
87 35 76 ... 77 38
```

Exercise09_08.java

(The Fan class) Design a class named Fan to represent a fan. The class contains:

- Three constants named SLOW, MEDIUM, and FAST with the values 1, 2, and 3 to denote the fan speed.
- A private int data field named speed that specifies the speed of the fan (the default is SLOW).
- A private boolean data field named on that specifies whether the fan is on (the default is false).
- A private double data field named radius that specifies the radius of the fan (the default is 5).
- A string data field named color that specifies the color of the fan (the default is white).

Exercise09_08.java

- The accessor and mutator methods for all four data fields.
- A no-arg constructor that creates a default fan.
- A method named toString() that returns a string description for the fan. If the fan is on, the method returns the fan speed, color, and radius in one combined string. If the fan is not on, the method returns the fan color and radius along with the string "fan is off" in one combined string.

Write a test program that creates two Fan objects. Assign maximum speed, radius 10, color yellow, and turn it on to the first object. Assign medium speed, radius 5, color blue, and turn it off to the second object. Display the objects by invoking their toString method. Here is a sample run of the program:

```
speed 3, color yellow, radius 10.0, fan is on  
speed 2, color blue, radius 5.0, fan is off
```


Exercise09_10.java

(Algebra: quadratic equations) Design a class named **QuadraticEquation** for a quadratic equation $ax^2 + bx + c = 0$. The class contains:

- Private data fields a, b, and c that represent three coefficients.
- A constructor with the arguments for a, b, and c.
- Three getter methods for a, b, and c.
- A method named getDiscriminant() that returns the discriminant ($b^2 - 4ac$)
- The methods named getRoot1() and getRoot2() for returning two roots of the equation

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

These methods are useful only if the discriminant is nonnegative. Let these methods return **Double.NaN** if the discriminant is negative.

Exercise09_10.java

Write a test program that prompts the user to enter values for a, b, and c and displays the result based on the discriminant. If the discriminant is positive, display the two roots. If the discriminant is 0, display the one root. Otherwise, display "The equation has no roots"

Here are some sample runs:

```
Enter a, b, c: 1 3 1 [Enter]
The roots are -0.3819660112501051 and -2.618033988749895
```

```
Enter a, b, c: 1 2 1 [Enter]
The root is -1.0
```

```
Enter a, b, c: 1 2 3 [Enter]
The equation has no roots
```

MyInteger.java

(The MyInteger class) Design a class named MyInteger. The class contains:

- An int data field named value that stores the int value represented by this object.
- A constructor that creates a MyInteger object for the specified int value.
- A getter method that returns the int value.
- The methods isEven(), isOdd(), and isPrime() that return true if the value in this object is even, odd, or prime, respectively.
- The static methods isEven(int), isOdd(int), and isPrime(int) that return true if the specified value is even, odd, or prime, respectively.

MyInteger.java

- The static methods `isEven(MyInteger)`, `isOdd(MyInteger)`, and `isPrime(MyInteger)` that return true if the specified value is even, odd, or prime, respectively.
- The methods `equals(int)` and `equals(MyInteger)` that return true if the value in this object is equal to the specified value.
- A static method `parseInt(char[])` that converts an array of numeric characters to an int value. For example, it should return 125 if the characters array is {'1', '2', '5'}.
- A static method `parseInt(String)` that converts a string into an int value.